BEING A PERFECT MORPHISM IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17875-17879 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17881-17898, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Dual of the ascending side already collected; the algebra core is a perfect-complex descent statement, faithfully flat down, not just stability.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Descent, lemma descending-properties-morphisms (the three-hypothesis criterion), Definition definition-perfect and lemma-flat-base-change-perfect of this chapter, and More on Algebra, lemma flat-descent-perfect for the algebraic core.

Source reference. Original lemma 0687: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17730–17736. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17898–17898. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0688: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17744–17746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 068T: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17898–17898. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is perfect" is fpqc local on the base.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. We will use the criterion of Descent, Lemma 02KP to prove this. By Definition 0687 being perfect is Zariski local on the base. By Lemma 0688 being perfect is preserved under flat base change. The final hypothesis (3) of Descent, Lemma 02KP translates into the following algebra statement: Let $A \to B$ be a faithfully flat ring map. Let $C = A[x_1, \ldots, x_n]/I$ be an A-algebra. If $C \otimes_A B$ is perfect as an $B[x_1, \ldots, x_n]$-module, then C is perfect as a $A[x_1, \ldots, x_n]$-module. This is More on Algebra, Lemma 068T. $\square$